

17 Ans: B

$$a_{\max} = \omega^2 x_o \text{ ----- (1) where } x_o = 0.3 \text{ cm}$$

$$\omega = \frac{2\pi}{T_{shm}} \text{ ----- (2)}$$

NB : T_{shm} = twice the period of $E_k - t$ graph [Remember this !]
 $= 2 \times 0.20 \text{ s} = 0.4 \text{ s}$